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Non-Contact Heart Rate Monitoring Using Multiple RGB Cameras

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Abstract. Recent advances in computer vision and signal processing are enabling researchers to realize mechanisms for the remote monitoring of vital signs. The remote measurement of vital signs, including heart rate (HR), Heart Rate Variability (HRV), and respiratory rate, presents important advantages for patients. For instance, continuous remote monitoring alleviates the discomfort due to skin irritation and/or mobility limitation associated with contact-based measurement techniques. Recently, several studies presented methods to measure HR and HRV by detecting the Blood Volume Pulse (BVP) from the human skin. They use a single camera to capture a visible segment of the skin such as face, hand, or foot to monitor the BVP. We propose a remote HR measurement algorithm that uses multiple cameras to capture the facial video recordings of still and moving subjects. Using Independent Component Analysis (ICA) as a Blind Source Separation (BSS) method, we isolate the physiological signals from noise in the RGB facial video recordings. With respect to the ECG measurement ground truth, the proposed method decreases the RMSE by 18% compared to the state-of-the-art in the subject movement condition. The proposed method achieves an RMSE of 1.43 bpm and 0.96 bpm in the stationary and movement conditions respectively.

Keywords: Photoplethysmogram, Independent Component Analysis (ICA), remote heart rate measurement, multiple camera monitoring.

1 INTRODUCTION

Vital signs, including HR, HRV, and respiration rate, are important indicators of the patient's health [1]. They provide critically needed information to make life-saving decisions. Continuous monitoring of HR and HRV is becoming an important aspect of home-based healthcare. In addition to physical health assessment, HR and HRV measurement can be employed for psychological health monitoring. For instance, mental stress can be effectively assessed using HR and HRV measures [2, 3]. Methods to measure HR are classified into two categories: Contact and Non-Contact based. Contact based measurement involves the use of sensors attached to the patient's skin. There are several limitations in contact based-measurement. These sensors limit the movement of the patient due to their wiring. Moreover, in some cases, connecting the sensor to the skin is not possible due to the patient's condition. For example, monitoring vital signs

in the Neonatal Intensive Care Unit (NICU) requires the use of adhesives to attach sensors to the skin of pre-term infants, which might result in pain and skin irritation [4]. However, non-contact measuring methods address effectively these issues. Optical non-contact methods estimate a photoplethysmography (PPG) signal without directly being placed on the subject's body. In recent studies, the PPG signals have been acquired using RGB (Red, Green, Blue) image sequences [5]. Furthermore, researchers have extracted HR[5], HRV [6] and respiration rate[7] parameters from the remotely measured PPG signals.

There are many approaches to recover the BVP. One of the most prevalent methods is BSS. Physiological signals are often mixed with other independent noise signals. Independent component analysis (ICA)[8] is a BSS technique that separates multivariate signals from underlying sources. Poh et al. [5] used ICA to extract the channel containing BVP information from the RGB channels of a facial video.

Ghanadian et al. [9] demonstrated the feasibility of monitoring HR during movement using a single camera. However, such system is limited to the Field of View (FoV) of a single camera. As soon as the subject steps outside of the camera's FoV, the physiological signal measurement is interrupted.

For the proposed method, we use a head pose detection scheme to ensure that we collect the best possible signal from the group of cameras interfacing to the system. The human head pose can be described by pitch, yaw and roll. Head pose detection has many applications in HCI research. For instance, the influence of banner advertisements on attention has been studied using head pose and gaze detection algorithms in [10]. Moreover, Stiefelhagen [11] used head pose detection to track the attention of the audience during meetings; they postulated that if an individual wants to pay attention to an object, he/she typically turns his/her head towards it.

In the proposed method, we prove that the application of multiple cameras in a room can increase the accuracy of remote HR estimation. There are many applications where the use of multiple cameras can refine the performance of remote physiological monitoring. For instance, in elderly care facilities, monitoring vital signs, especially HR, can be achieved more accurately using multiple cameras mounted at each corner of the rooms. Moreover, in some cases, existing surveillance cameras can double as remote physiological monitoring sensors.

2 BACKGROUND AND RELATED WORKS

Nanfei et al. [12] presented the first remote pulse measurement scheme using thermal imaging. They achieved the HR measurement by computing the frequency of pulses obtained from changes in the temperature of the vessels. The temperature is regulated by pulsatile blood flow. However, due to the thermal sensitivity of the sensor, the changes in the room temperature can have an adverse effect on the performance of the system [13]. Moreover, this approach requires the use of a relatively expensive thermal camera compared to RGB cameras.

Takano and Ohta [14] measured the cardiac pulse rate using a digital camera. They calculated the average brightness of time-lapse skin images recorded by a CCD camera. They extracted the HR using auto-regressive spectral analysis on the video frames.

McDuff et al. [15] developed an HR and HRV measurement system that uses a five band camera (Red, Orange, Green, Cyan and Blue) to capture the facial video recording. To find the facial landmark coordinates that allow them to localize the ROI, they utilized the local evidence aggregated regression algorithm. They presented the spatially averaged color spaces to the ICA to separate the source signals from noise. Their results showed that combining the Cyan, Green and Orange channels achieved better results than any other color combination. Similar to thermal cameras, the five band cameras are comparatively expensive.

Poh. et al. [5] proposed to recover the HR using RGB cameras. They used the openCv [16] face detection method to locate the face, and a face tracker algorithm to track the head motions of the subjects during measurement. The movements of subjects were restricted to small motions of the head such as nodding, looking up/down, and leaning forward/backward.

Ghanadian et al. [9] investigated the possibility of HR measurement in the presence of movement. They utilized a low cost RGB camera to record facial images. Then, they presented the spatially averaged color channels over the images to the ICA. They proposed a machine-learning algorithm to choose the component generated through ICA that is most reflective of PPG. Moreover, they presented an algorithm to equalize the light effects on the facial images by normalizing the light across the frames of video segments. They posited that 50% of the face of the subjects should be visible to the camera to make accurate measurements.

McDuff [17] estimated the HRV of stationary subjects using multiple cameras. They evaluated their method under various levels of motion while the subjects were seated. Their method still has some limitations. The subjects cannot roam the room freely.

Other works that considered multiple cameras to extract vital signs remotely [18][19] used an infra-red and an RGB camera to capture two PPG signals at different wavelengths. However, our method uses multiple cameras to increase the FoV resulting from the combination of the images simultaneously captured by all cameras.

3 METHOD

We base our approach on the system proposed by Ghanadian et al.[9] Fig. 1a depicts the overall processing steps involved in the latter method. They used a single camera to detect the face of still and moving subjects. The movement of the subjects was limited to walking towards and away from the camera. They required the capture of at least 50% of the face as ROI to extract the PPG signal. To reduce the negative effects of light changes as the subject roamed the room, they decomposed the ROI to Hue, Saturation and lightness (HSL) color space and normalized the L value. Then, they decomposed the ROI to extract the RGB signals by averaging the Red, Green and Blue values over the ROI pixels in each frames. They presented the RGB signals to the ICA to separate them into three independent components. They proposed a machine Learning (ML)

algorithm to select the best signal generated through ICA that reflects PPG. They calculated the HR from the PPG signal.

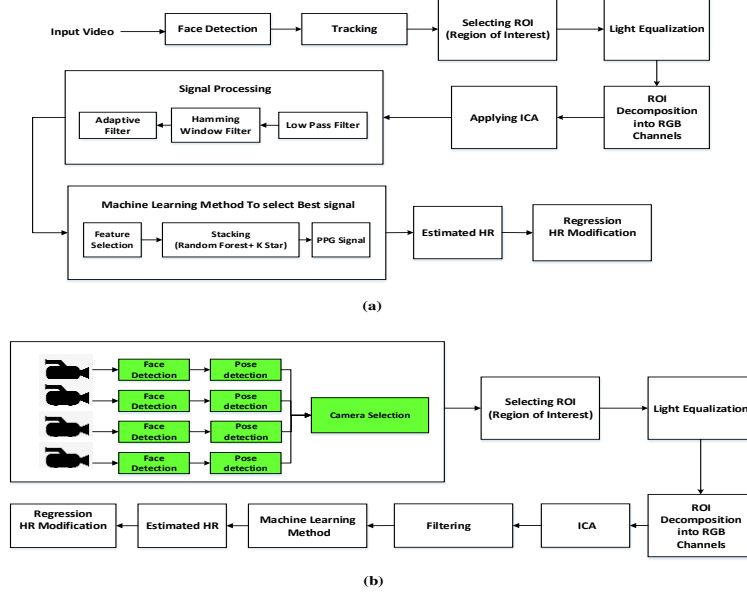


Fig. 1. HR prediction algorithm for the (a) Ghanadian et al. [9] and (b) proposed method

Fig. 1b shows the steps necessary to extract HR from video using the multi-camera proposed approach. In the following sub-sections, we describe the new modules added to the method described in [9].

3.1 Face and Head Pose Detection

During movement, the angle of the subject's face with respect to the camera changes frequently. Hence, we utilize multiple cameras to capture the subject's face under a variety of head poses. We use a face detection method to capture the face in a video recorded by a standard RGB camera. We utilize four 24-bit RGB cameras recording at 30-frames per second and 640×480 pixels per frame. Each camera operates independently and simultaneously with other cameras. We locate the face in all of the video streams (originating from all the cameras) using a face detection method (OpenFace) [16] and calculate the head pose of the subject with respect to each camera. OpenFace [16] identifies sixty-eight facial landmarks including points on the nose, eyebrows, chin, eye corners and mouth for face detection and pose estimation [20].

3.2 Camera Selection

We assume there are N cameras in the room. Hence, we divide the room into ω non-intersecting partitions, where ω is calculated using equation (1). We place the cameras

along the perimeter of a circle co-centered with the room. We ensure that the distance between adjacent cameras is equal for all cameras.

$$\omega = \frac{360^\circ}{N} \quad (1)$$

The system continuously selects every time-period t the most appropriate camera to measure the HR. The head pose angle of the subject is calculated with respect to each camera's axis. We refer to this angle as A_i where i is the index of the camera and ranges from 0 to $N-1$. Hence, the head pose angle of a subject looking directly at the camera is 0 degrees. Fig. 2 shows the different head pose degrees with respect to each camera. To calculate HR, for time-period t , we use the frames captured by the camera corresponding to the smallest A_i . Hence, we create a series of images originating from one or more cameras. This series of images is processed in the same way a video captured from a single camera is treated by the algorithm presented in [9]. Hence, the light equalization method of [9] is applied to this series of images, which reduces the effect of light differences between the various frames originating from the camera(s) in an unevenly lit room.

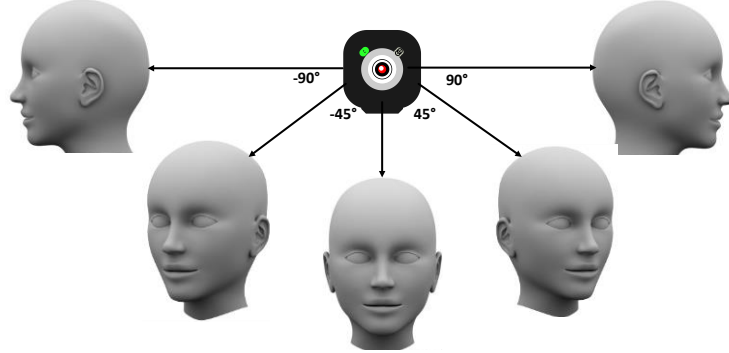


Fig. 2. Head Pose degrees with respect to a camera

For the experimental scenario, we employed four cameras placed at each corner of a square-shaped room. Therefore, video frames depicting a subject's head pose angle between $[-45^\circ, 45^\circ]$ are used to estimate the HR. If we assume an HR monitoring session lasts M measurement time-periods, hence:

$$T = M \times t \quad (2)$$

Where T is the length of the recording session. Therefore, to measure HR during T , we collect M sets of frames from the N available cameras. The M sets of frames are considered the input video to the HR estimation method.

We select the entire face of the subjects as the ROI to increase the amount of information extracted from the face. By increasing the distance of the subjects from the cameras, the ROI becomes smaller, hence choosing the entire face allows us to maximize the useful information. To construct the raw RGB signals, the selected ROI in every frame is spatially averaged over all the pixels. Fig. 3 shows an example of how the frames are compiled from N cameras to produce a single facial video.

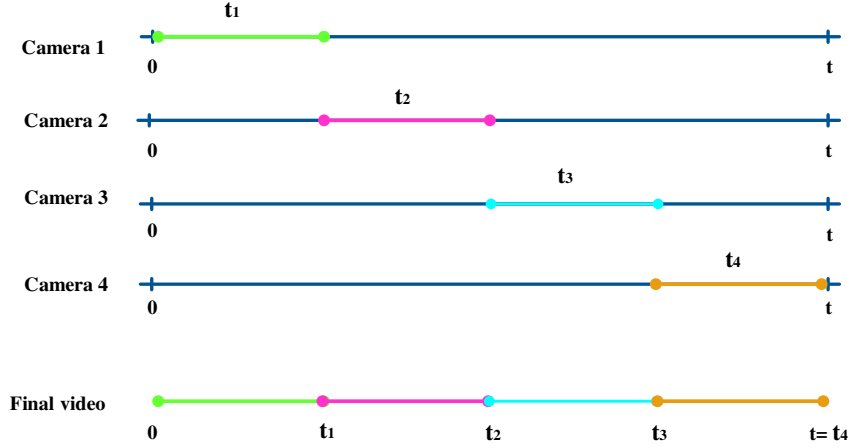


Fig. 3. Face images extraction from multiple camera in stationary mode

Using multiple camera increases the probability of capturing the face with one of the cameras. However, evidently, there are no guarantees that one of the cameras captures the face at an appropriate angle. For instance, the subject might come very close to the camera which crops her/his face out of the frame.

4 DATASET

The experiments discussed in this section were approved by the Office of Research Ethics and Integrity at the University of Ottawa (Certificate Number: H02-17-13). We collected a dataset with ten volunteering subjects (3 female and 7 male) who signed consent forms to participate in this study. The dataset consists of two subsets: stationary and movement mode subsets.

4.1 Stationary Mode

We asked participants to sit in front of one camera and rotate towards the next camera with a speed of 30 degrees/second. We asked them to stop for 10 second while facing each camera with minimum movement. The subjects wore a physiological sensor

(Zephyr Bioharness¹) that collects an ECG signal at a sampling rate of 250 Hz. The data collected by the sensor is used as ground truth to evaluate the accuracy of our results. Fig.4a shows the experimental setup for the stationary mode using multiple cameras. The videos were collected by four conventional RGB camera (Logitech Webcam C920) filming at 30-frames per second and 640×480 pixels per frame. The cameras were mounted on the four corners of a 4×4 meter square-shaped room and the subject was seated at the center of the room. The lighting in the room was a mixture of natural sunlight coming from windows and fluorescent ceiling light. Table 1 describes the conditions for the stationary mode subset collection with multiple cameras.

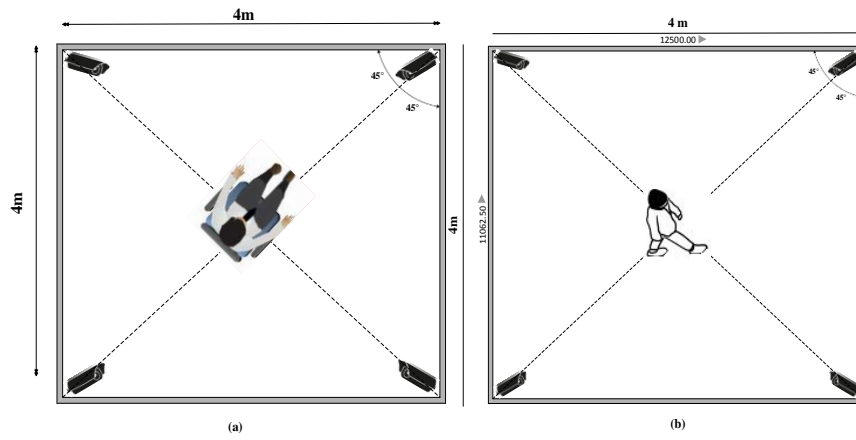


Fig. 4. Experimental setup for (a) Stationary mode, (b) Movement mode

Table 1. Description of the stationary and movement mode data subsets

Parameters	Movement Mode	Stationary Mode
Experiment length per subject (second)	240	180
# of participants	10	10
	(3 F, 7 M)	(3 F, 7 M)
Distance from camera (meter)	0-4	2
Recording rate (fps)	30	30
# of 1 min. video segments	40	30
# of frames	72000	54000

4.2 Movement Mode

For the Movement Mode subset, we asked the subjects to move in the room for 4 minutes by walking, talking, or lying down (see Fig. 4b). The dimensions of the room

¹ “Zephyr TM Performance Systems | Performance Monitoring Technology.” [Online]. Available: <https://www.zephyranywhere.com/>. [Accessed: 17-Apr-2018].

are 4m×4m. The subjects could move freely without any constraint. The rest of the experimental conditions, including lighting, camera, and sensor were identical to the ones used for the stationary mode. Table 1 shows the conditions for the movement mode subset collection with multiple cameras.

5 EVALUATION

In this section, we present the evaluation of the proposed contributions as detailed in Section 3. The dataset used in the experiments is described in Sections 4.1 and 4.2.

5.1 Stationary Mode

We compare the proposed approach to the methods of Poh et al. [5] and Ghanadian et al. [9] in the stationary mode. Both of these methods used a single camera. The proposed method decrease the RMSE by 0.04 and 0.95 compared to Ghanadian et al. [9] and Poh et al. [5] respectively.

Table 2. Comparison between Poh et al.[5], Ghanadian et al.[9], and the proposed method

Parameters	Poh et al. [5]	Ghanadian et al. [9]	Proposed method
Mean Bias (bpm)	-0.12	0.78	0.74
SD of Bias (bpm)	3.11	1.32	1.29
Upper limit (bpm)	6.09	2.58	2.53
Lower limit (bpm)	-6.33	-1.03	-1.79
RMSE (bpm)	2.38	1.47	1.43
Corr. coefficient	0.98*	0.98*	0.99*

*: (p<0.004)

5.2 Movement Mode

Ghanadian et al. [9] utilized a single camera to capture the facial videos. The level of movement in these videos is limited to walking towards and away from the camera. In the proposed method, the subjects can move freely in the room. The proposed method can fail to capture the physiological signal only if none of the cameras can capture the face. We test the Ghanadian et al. [9] method in a scenario where subjects can roam the room without any restrictions (using the dataset described in Section 4.2). Table 3 presents a comparison between the proposed method and Ghanadian et al.[9] in the movement mode. The use of multiple cameras drastically decreases the RMSE from 5.08 bpm to 0.96 bpm.

Table 3. Comparison of Ghanadian et al.[9], and proposed method using multiple cameras in movement

Parameters	Ghanadian et al. [9]	Proposed method
Mean Bias (bpm)	6.65	1.08
SD of Bias (bpm)	6.34	1.36
Upper limit (bpm)	12.42	2.66
Lower limit (bpm)	-5.77	-1.58
RMSE (bpm)	5.08	0.96
Corr. coefficient	0.85*	0.98*

*: (p<0.004)

Fig. 5 presents the performance of the proposed method and bland Altman plot. Bland-Altman plot in Fig.5 analyze the agreement between the estimated HR from the proposed algorithm using multiple cameras and the HR extracted from the ground truth (i.e. HR obtained from the Zephyr Bioharness ECG sensor).

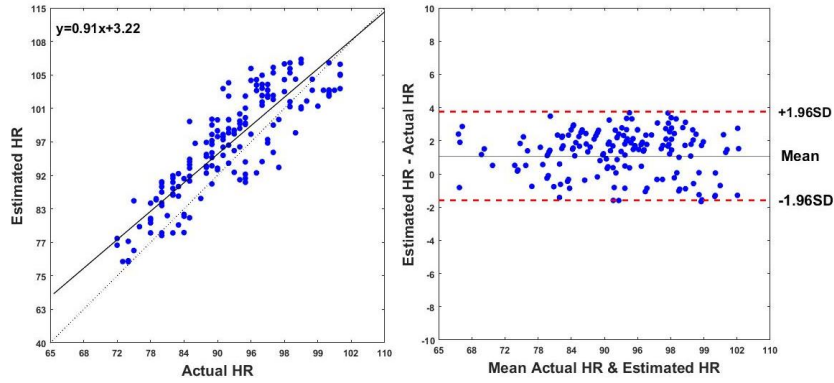


Fig. 5. Statistical Results of Bland and Altman plot In Multiple camera experiment

6 DISCUSSION

The results show that using multiple cameras can improve the accuracy of remote HR estimation. The combination of multiple cameras creates a larger collective FoV which gives the patient more freedom in the movement during monitoring without risking interruptions. Each frame's ROI is spatially averaged and corresponds to a data point

in the remotely measured PPG signal. During movement, the subject's ROI might fall out of the FoV in a single camera solution thus degrading the quality of the PPG signal which in turn adversely affects the accuracy of the HR estimate.

Collecting frames from several cameras produces a series of images taken from various vantage points which results in a variety of lighting conditions. Therefore, equalizing the lightness across the frames stabilizes the lighting condition.

7 CONCLUSION

Remote monitoring of HR in real-life situations requires a robust technology that can measure continuously with the fewest possible number of interruptions. We realize that using a single camera limits the FoV of the system. Hence, we proposed a remote HR monitoring method that uses multiple cameras to address the latter limitation. The proposed method presents the possibility of continuous measurement of physiological vital signs for still and moving subjects. To realize a multi-camera system, we combine video segments from multiple cameras. We achieved an RMSE of 0.96 bpm for the estimation of HR during subject movement.

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